User Manual Document

Cognizant Framework for GenAI induced Efficiency Calculation

8th September 2024

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1. Introduction

1.1. Purpose

The purpose of this user manual is to provide comprehensive guidance for users of the Gen-AI Led Efficiency Calculator. This document aims to:

1. **Facilitate Understanding**: Offer clear explanations of the tool's features, functionalities, and benefits, ensuring users can effectively navigate and utilize the application.
2. **Provide Step-by-Step Instructions**: Deliver detailed, step-by-step instructions on how to configure the calculator, input data, and interpret results, helping users maximize the tool's potential.
3. **Aid in RFP Processes**: Provide insights on how the application framework can be effectively utilized in Request for Proposals (RFPs), helping account teams articulate their AI strategy and expected efficiencies.
4. **Support Efficient Implementation**: Assist account/project teams in integrating the calculator into their project workflows, promoting efficient use of AI solutions for enhanced productivity.

1.2. Overview of the Solution

The Gen-AI Led Efficiency Calculator is an innovative tool designed to calculate the overall efficiency gain for implementing artificial intelligence solutions. This solution empowers users to assess and quantify the potential efficiency gains from implementing various AI technologies in their Software Testing Life Cycle (STLC) activities.

The calculator framework estimates the reduction in effort/resources year-on-year, based on the selected AI solutions tailored to specific STLC phases. Users can configure a range of predictive and generative AI options, each with defined efficiency gain percentages, to reflect their unique project needs.

The application not only simplifies the decision-making process for integrating AI into projects but also provides a clear visualization of the long-term benefits. As users input their project details and select relevant AI solutions, the calculator dynamically computes the overall efficiency gains, helping teams understand the impact of AI implementation on their workflows.

1.3. Target Audience

This user manual is intended for members teams who are involved in software testing and quality assurance processes and plans to leverage AI solutions in their projects. It is particularly beneficial for:

1. **Proposal Writers**: They can use the calculator to demonstrate potential efficiency gains from AI solutions, making their proposals more compelling.
2. **Business Development Teams**: Teams responsible for responding to RFPs can leverage the calculator to highlight the benefits of AI integration, aligning with client needs for improved efficiency.
3. **Technical Consultants**: Consultants can use the tool to provide data-driven insights into how AI can enhance project outcomes, strengthening their RFP responses.
4. **Project Managers**: They can estimate efforts savings and efficiency improvements when drafting project plans in response to RFPs.

2. Getting Started

The solution is currently hosted on the Cognizant server and can be access using the following URL: http://pirategic.eastasia.cloudapp.azure.com:280/login. To gain access to this URL, please reach out to our support team via the email provided in the **Contact information for customer support:** section of this document.

2.1 Initial Solution Setup Guide

As part of the initial setup, users should navigate to the Configuration page. Here, they need to select the appropriate project category and review the default STLC effort distribution percentages found under the "STLC Effort" tab, as detailed in section **3.1 Configuration** guide. The percentage distributions for the three different project categories have been validated by the Tech COE team. However, users have the flexibility to modify the percentage allocations for various STLC activities on the "STLC Effort" tab. If a particular STLC activity is not applicable to the selected project category, users can set its percentage to zero.

It's important to ensure that the total percentage across all STLC activities for a specific project category always equals 100%.

Additionally, on the AI Solution Catalogue page, default values for effort savings associated with each AI solution have been established in consultation with the Tech COE team. Users can override these default savings or percentage distributions, if necessary, as discussed in section **3.1 Configuration** guide.

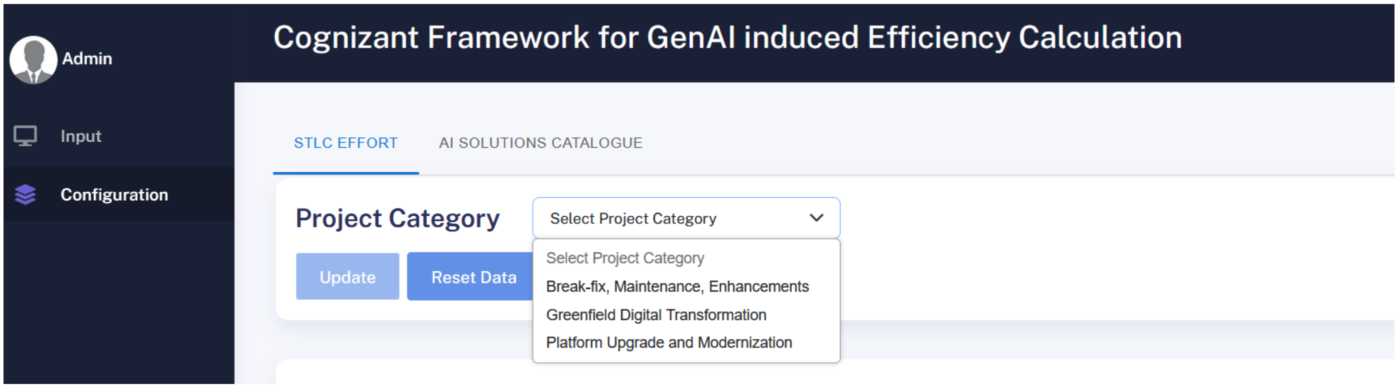
3. Solution Features and Functionalities

There are mainly 2 modules in GenAI induced Efficiency Calculator solution which given below:

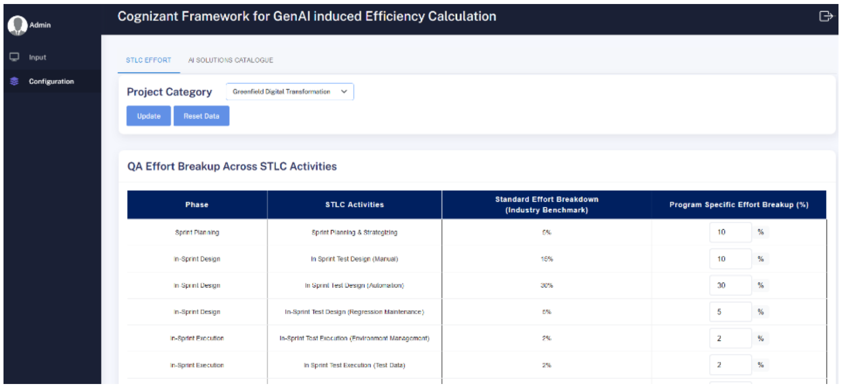
3.1 Configuration

Configuration module can be used for maintaining STLC phase efforts and AI Solution Catalogue. **Configuration** module comprises of primarily 2 submodules: 1) **STLC Efforts** 2) **AI Solution Catalogue.**

1) **STLC Efforts:** User can maintain the % efforts break-up across the various STLC activities from this page. There are 3 types of project category as show in the below screen:
   - Break-fix, Maintenance, Enhancements
   - Greenfield Digital Transformation
   - Platform Upgrade and Modernization



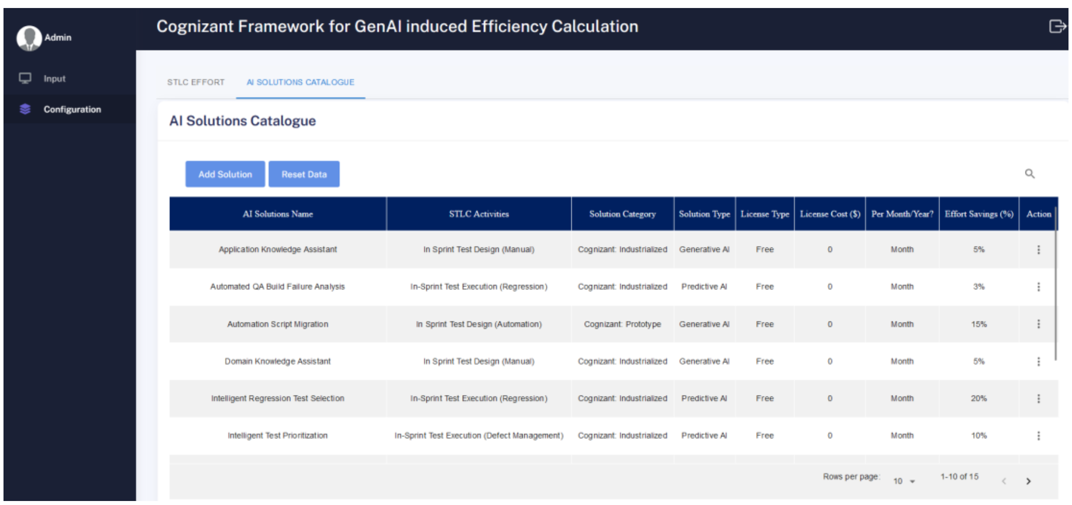
Upon selecting any project category the default QA Effort breakup % will be displayed, as given below:



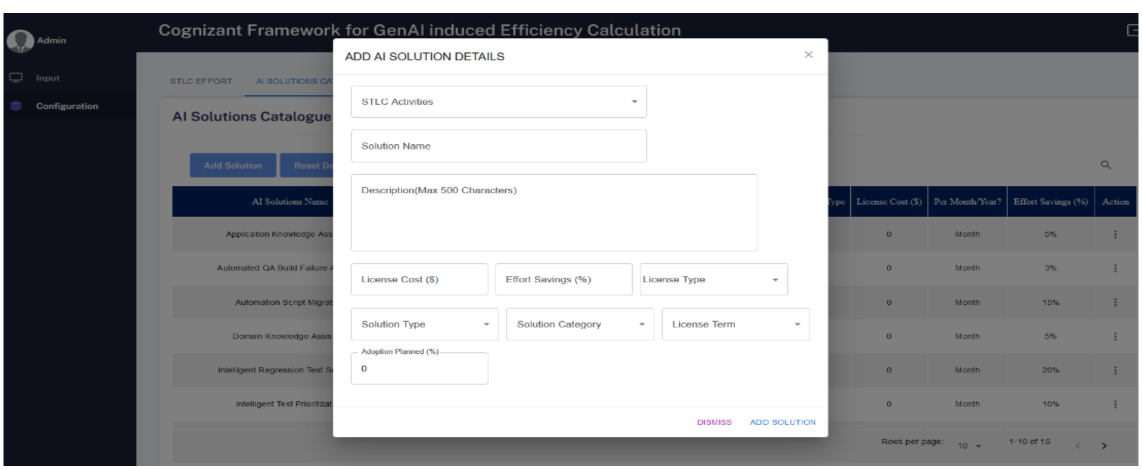
Users can review the standard effort breakdown (industry benchmark) in the designated column and can adjust any percentage distribution in the "Program Specific Effort Breakup" column. All values are expressed as percentages, so the total of all entries in the "Program Specific Effort Breakup" column must equal **100%**. Users may set any percentage to zero (0) if desired.

If users wish to revert to the industry benchmark for all values, they can simply click the "Reset Data" button.

2) **AI Solutions Catalogue:** Users can add and manage all solutions from this tab. Upon loading, the tab displays all previously added solutions as shown below.

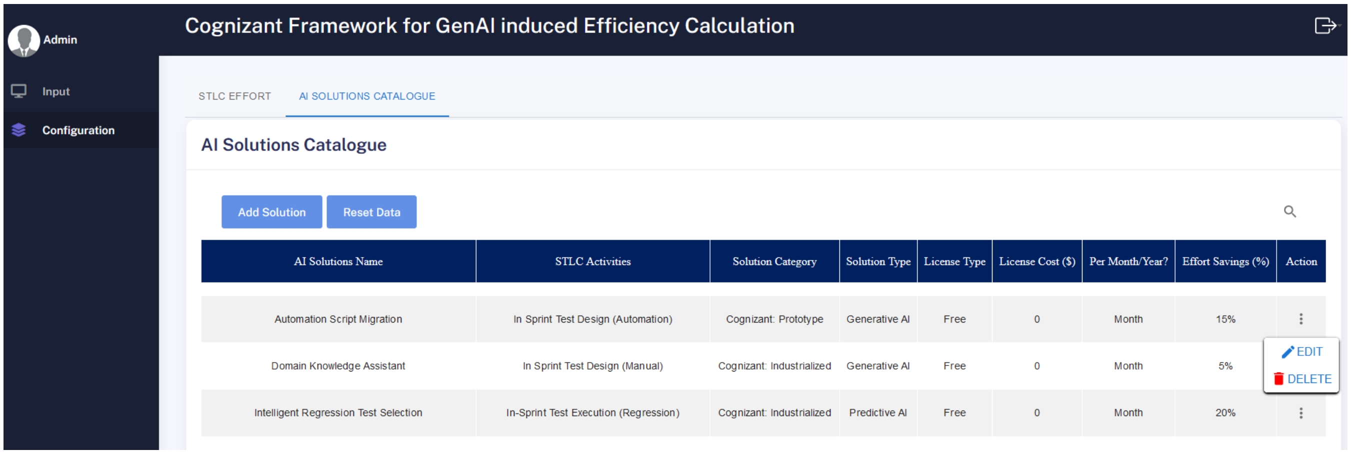


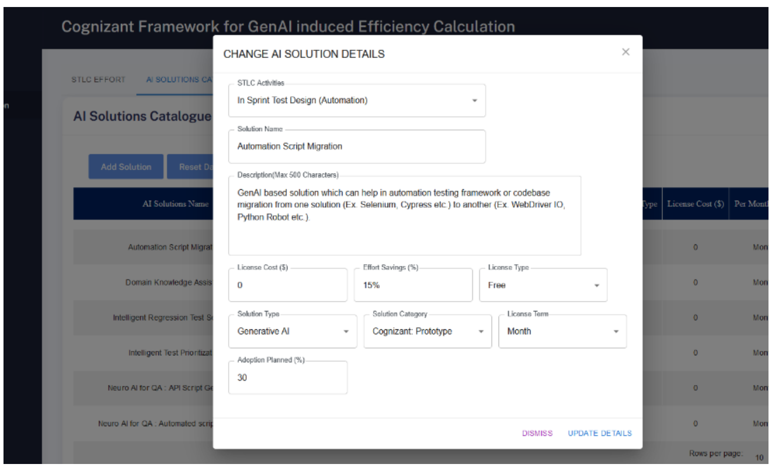
To add a new solution, users should click the "Add Solution" button, which will open a popup window. Users must fill in all required information on this screen and then click the "Add Solution" button to include the solution in the list.



User needs to fill all data from this screen and then have to click on "Add Solution" button to add this solution to the list.

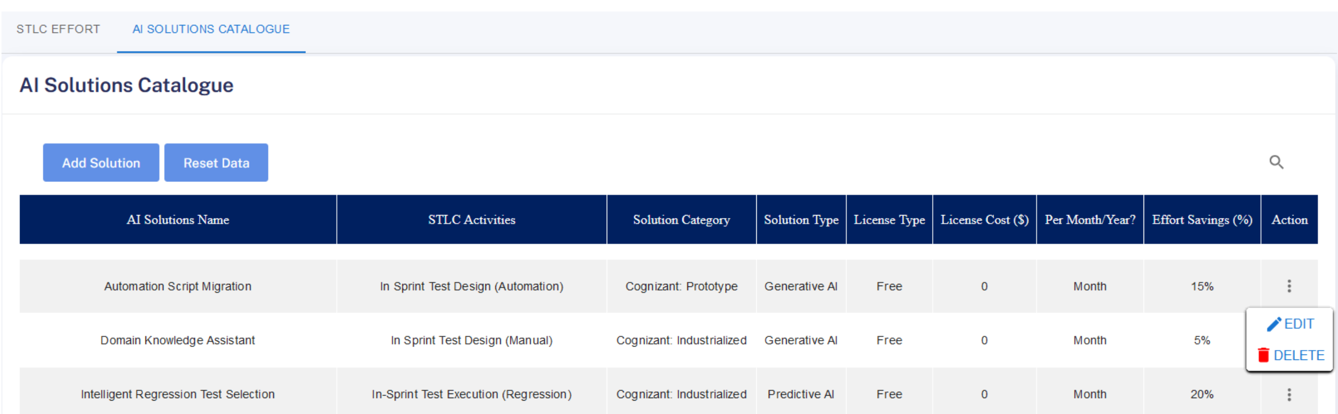
Additionally, there is an option to edit existing solutions. To do this, users can click on the three dots in the status column next to the specific solution they wish to edit, as illustrated in the image below.





To edit a solution, users should update all necessary information and then click the "Update Details" button.

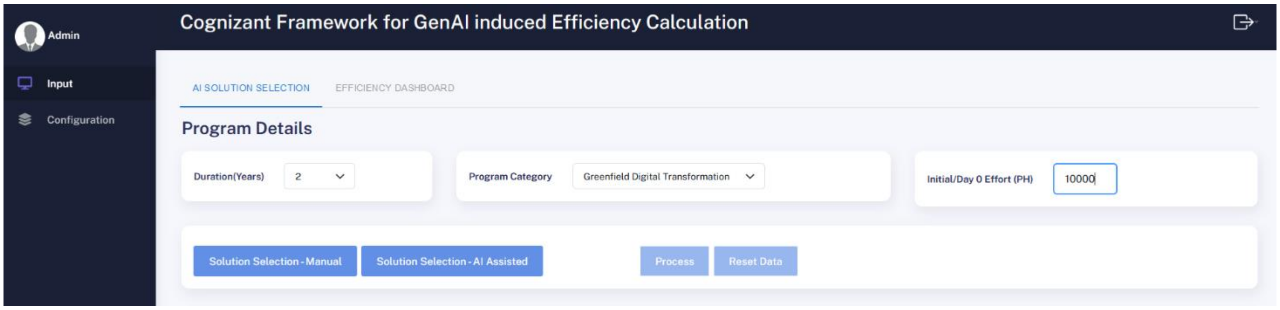
To delete a specific solution, users need to click on the three dots in the status column and select "Delete" from the menu, as shown in the image below.



The "Reset Data" button can be used to restore all data to its default settings.

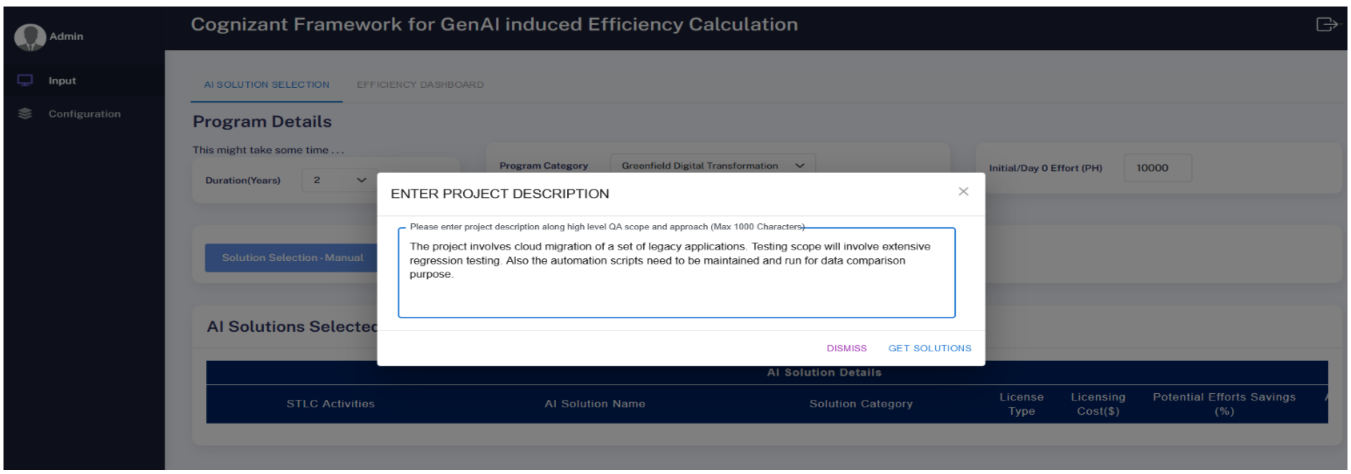
3.2 Input:

The input module allows users to enter data for efficiency calculations. To activate the “Solution Selection” button, users must select the duration, program category, and initial effort, as illustrated in the image.

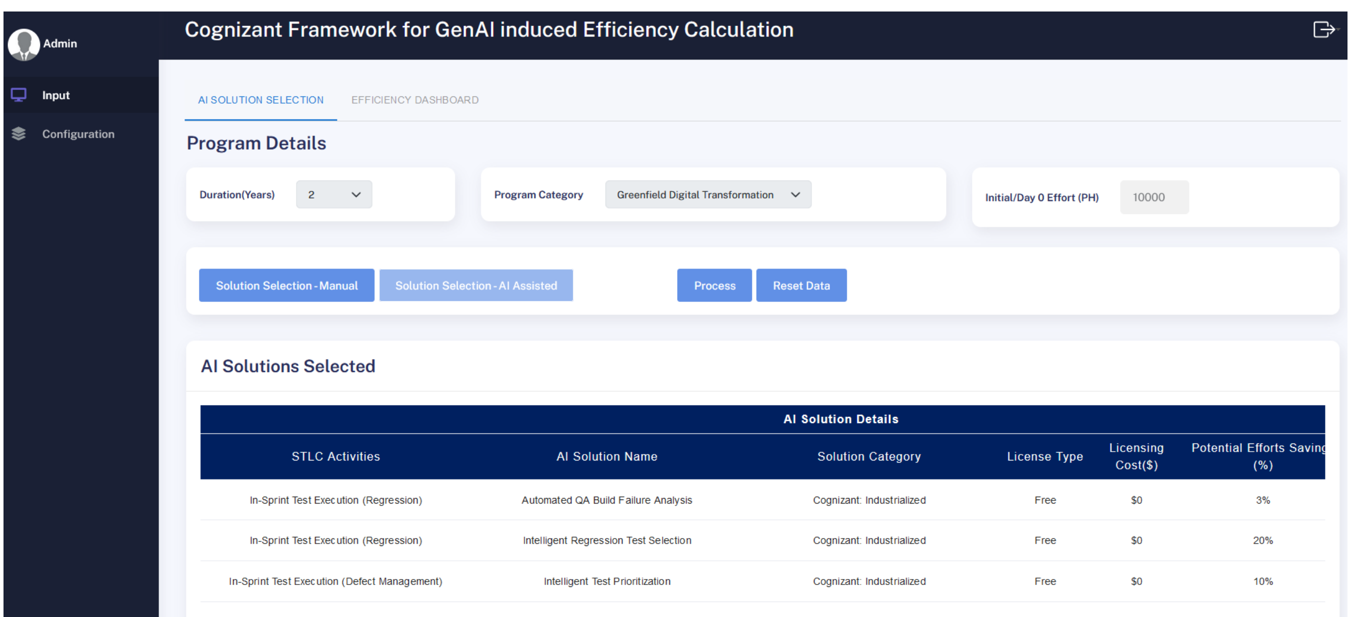


Once the details are provided on the previous screen, the solution selection will be enabled. Users can manually select a solution or receive automatic suggestions from Gen-AI.

To select a solution using AI assistance, users must click the “Solution Selection - AI Assisted” button and provide a brief description of the project, limited to 1000 characters, as shown below.



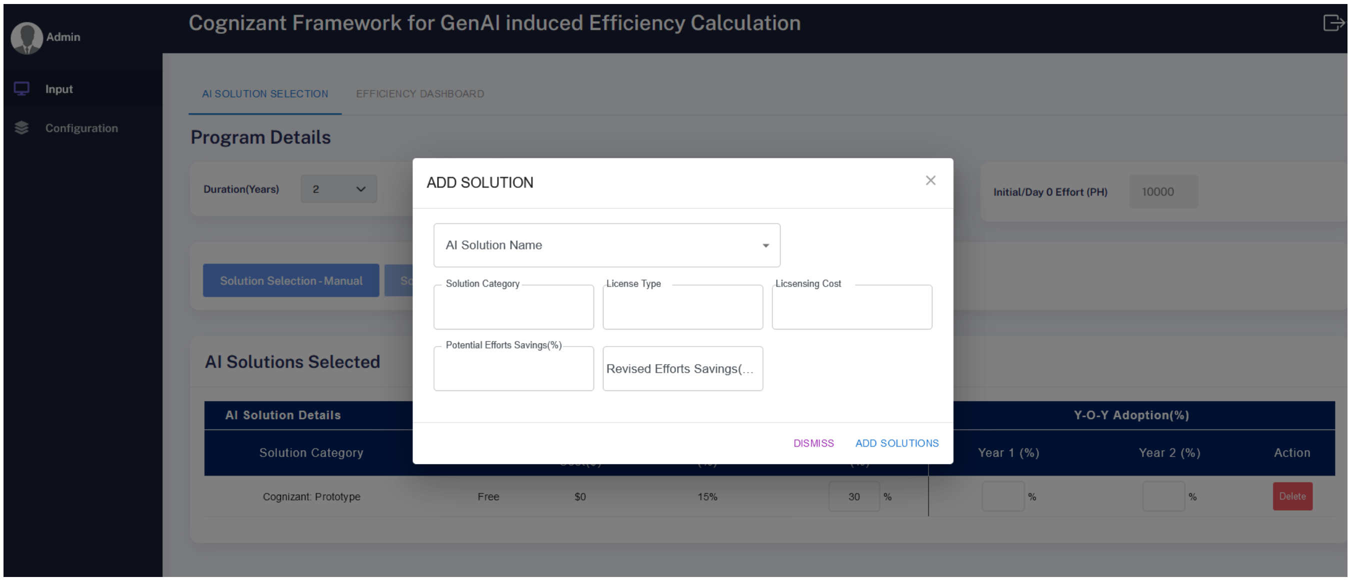
After entering the project details, users need to click the “Get Solution” button to display a list of all solutions selected through AI tools, as shown below.



Once the solutions are listed, the “Solution Selection - AI Assisted” button will be disabled for further use. However, users can remove any solution from the list using the delete button located on the right side.

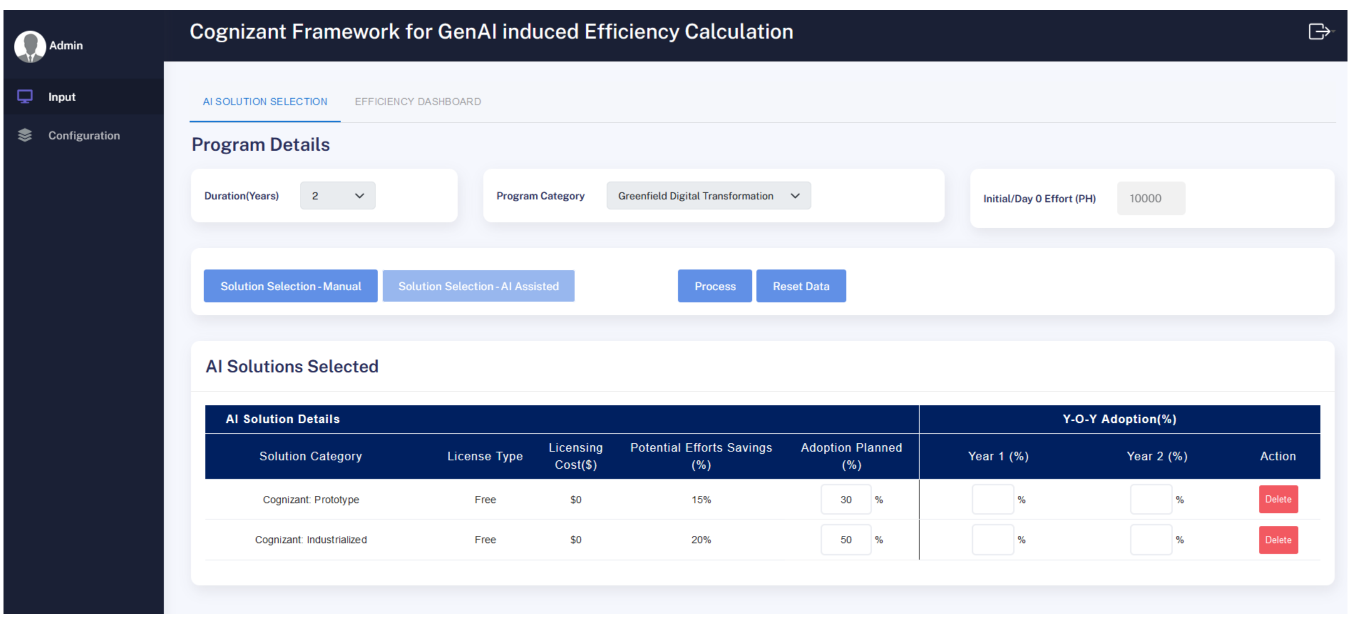
Additionally, there is an option to add more solutions through “Solution Selection – Manual.”

To manually add a solution, users should click the “Solution Selection – Manual” button and select their preferred solution, as shown in the image.



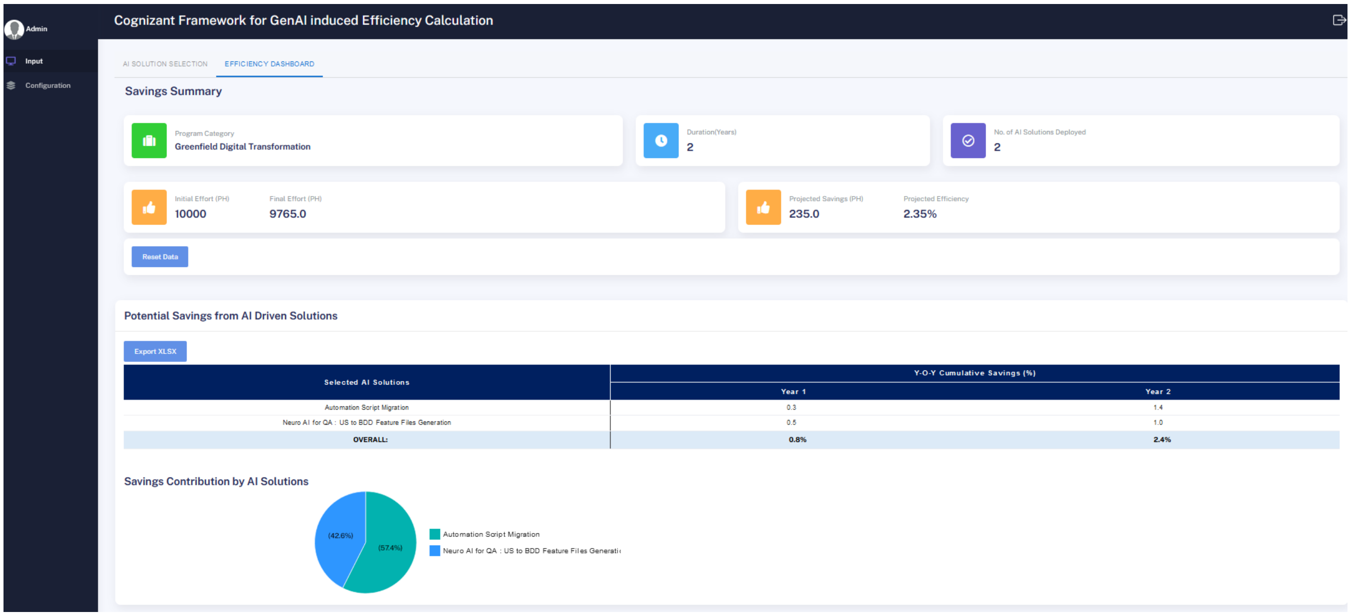
Note: If a user selects a solution manually, the “Solution Selection - AI Assisted” button will be disabled for further use.

Once all preferred solutions are selected, they will be listed under the AI Solution Selected table. Users can review the adoption percentages, which will be prepopulated in this table, as shown below.

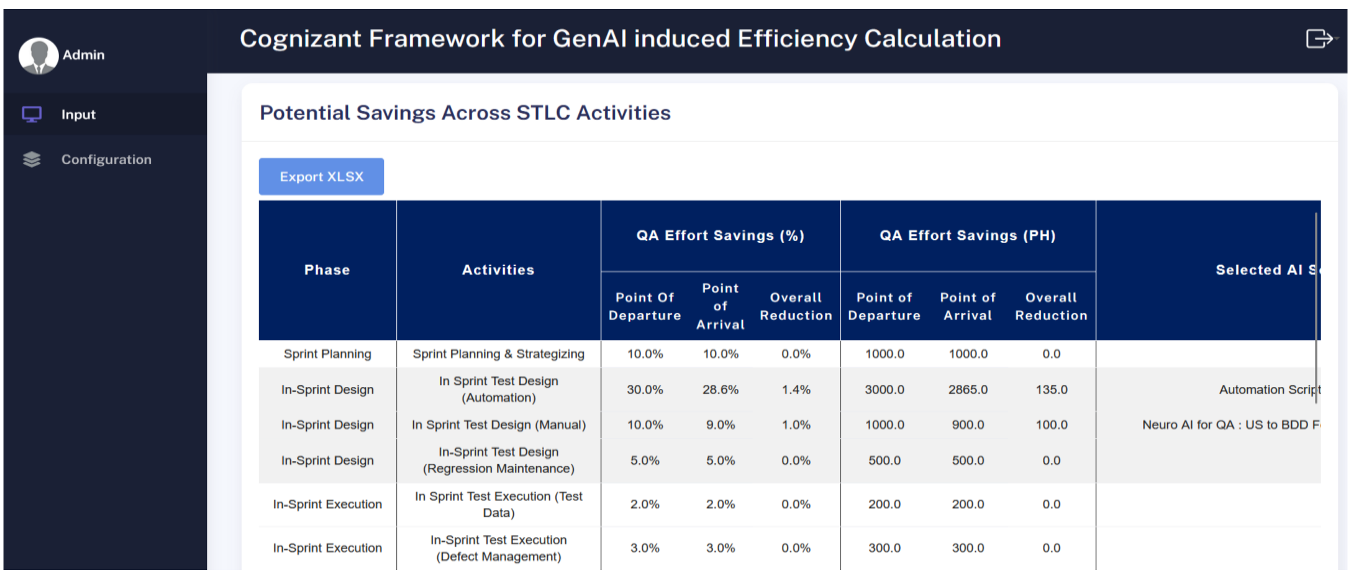


Users can adjust the adoption percentages as needed. Additionally, they must enter the year-over-year (Y-O-Y) adoption percentages, for example, 50% for Year 1 and 100% for Year 2. The adoption percentage for Year 1 must be lower than that for Year 2, and Year 2 must be lower than Year 3, and so on. Moreover, the adoption percentage for the final year (Year 2 in this case) must always be set to 100%, as full implementation is expected to be completed by that year. After filling all data then user need to click on “Process” button to generate the efficiency calculation as below.

The following image displays the year-over-year (Y-O-Y) effort gains aggregated and calculated at the selected AI solutions level.



Wheras, the following image illustrates the Y-O-Y effort gains aggregated and calculated at the STLC phase and activity levels, including the Points of Arrival and Departure, along with the overall effort reduction for each STLC phase.



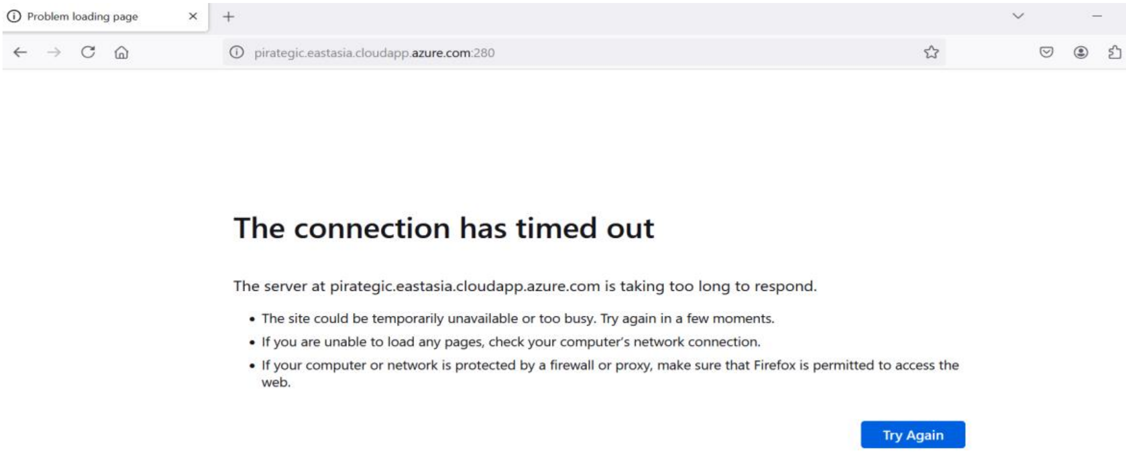
Users can clear all data at any time using the “Reset Data” button. Additionally, they can utilize the “Export XLSX” feature to export the table data to Excel.

4. Troubleshooting

Following is some of the common issues which may occur during the time of initial data setup or while using the solution:

4.1 Common Issues & Solutions:

1) Sometimes the users may run into the following issue:



Solution: To fix this issue user need to first check their internet connection and URL used to access the Tool, if everything is found to be okay then user should contact the Tool support team via email provided in the Contact Information for Customer Support given in this document below.

4.2 Frequently Asked Questions

1. **What is the Gen-AI Led Efficiency Calculator?**
    - The Gen-AI Led Efficiency Calculator is a tool designed to calculate Y-O-Y potential efficiency gains from implementing AI solutions in the Software Testing Life Cycle (STLC).
2. **How do I start using the calculator?**
    - Begin by selecting your project category and entering the initial effort. This will activate the solution selection options.
3. **What types of project categories are available?**
    - There are three main project categories available in the STLC Efforts section 1) Break-fix, Maintenance, Enhancements 2) Greenfield Digital Transformation 3) Platform Upgrade and Modernization each displaying a QA effort breakdown.
4. **Can I modify the standard effort breakdown?**
    - Yes, you can override the industry benchmark values in the "Program Specific Effort Breakup" column. Ensure that the total sums to 100%.
5. **What happens if I click the "Reset Data" button?**
    - Clicking "Reset Data" will revert all values back to the industry benchmark for the selected project category.
6. **How can I add a new AI solution?**
    - Click on the "Add Solution" button in the AI Solutions Catalogue tab, fill out the required fields in the popup, and click "Add Solution."
7. **Can I edit or delete existing AI solutions?**

- Yes, you can edit a solution by clicking the three dots under the status column and selecting "Edit," or delete it by choosing "Delete" from the same menu.

8. **What is the purpose of the Input module?**
   - The Input module allows users to enter data for efficiency calculations, including duration, program category, and initial effort.
9. **How does the AI-assisted solution selection work?**
   - After providing project details within 1000 characters, click "Get Solution" to receive AI-suggested solutions tailored to your project.
10. **Can I select solutions manually after using AI-assisted selection?**
    - No, once you use the AI-assisted selection, that option will be disabled. You can only add solutions manually if you haven't used AI selection.

5. Supports and Resources

5.1 Contact information for customer support:

For any inquiries or assistance regarding the Cognizant Framework for GenAI induced Efficiency Calculation, please reach out to our support team at the below DL.

Cognizant Support Team Contact Details:

INSQEAGENAICommunity@cognizant.com

6 Appendices

6.1 Glossary Of Terms

1. **AI (Artificial Intelligence)**: Technology that simulates human intelligence processes, including learning, reasoning, and self-correction.
2. **STLC (Software Testing Life Cycle)**: A series of phases that define the process of software testing, from planning to execution and closure.
3. **Efficiency Gain**: The improvement in productivity or reduction in resource usage achieved through the implementation of AI solutions.
4. **Program Category**: Different types of projects or programs within the STLC that can be selected for analysis.
5. **QA (Quality Assurance)**: A systematic process to ensure that software products meet specified requirements and standards.
6. **Effort Breakdown**: The distribution of resources and time allocated to various activities within a project.

7. **Industry Benchmark**: Standardized metrics used to compare performance or efficiency against similar projects or practices in the industry.

8. **Program Specific Effort Breakup**: Customizable percentages reflecting the effort distribution specific to a particular project or program.

9. **AI Solution Catalogue**: A repository of AI technologies and tools available for selection and integration into the testing process.

10. **RFP (Request for Proposal)**: A document issued by a company inviting suppliers to submit proposals for a specific project or service.

11. **Predictive AI**: AI technologies that analyze data to predict future outcomes or trends.

12. **Generative AI**: AI technologies capable of generating new content or solutions based on existing data.